MICROECONOMICS OF COMPETITIVENESS · GROUP 1

Energy Cluster Norway

From inherited to created prosperity

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Norway must turn the prosperity it *inherited* from petroleum into prosperity it **creates** with globally competitive energy technologies.

1

Diagnosis

Performance, business environment and policy

2

Cluster

History, size, actors and Diamond

3

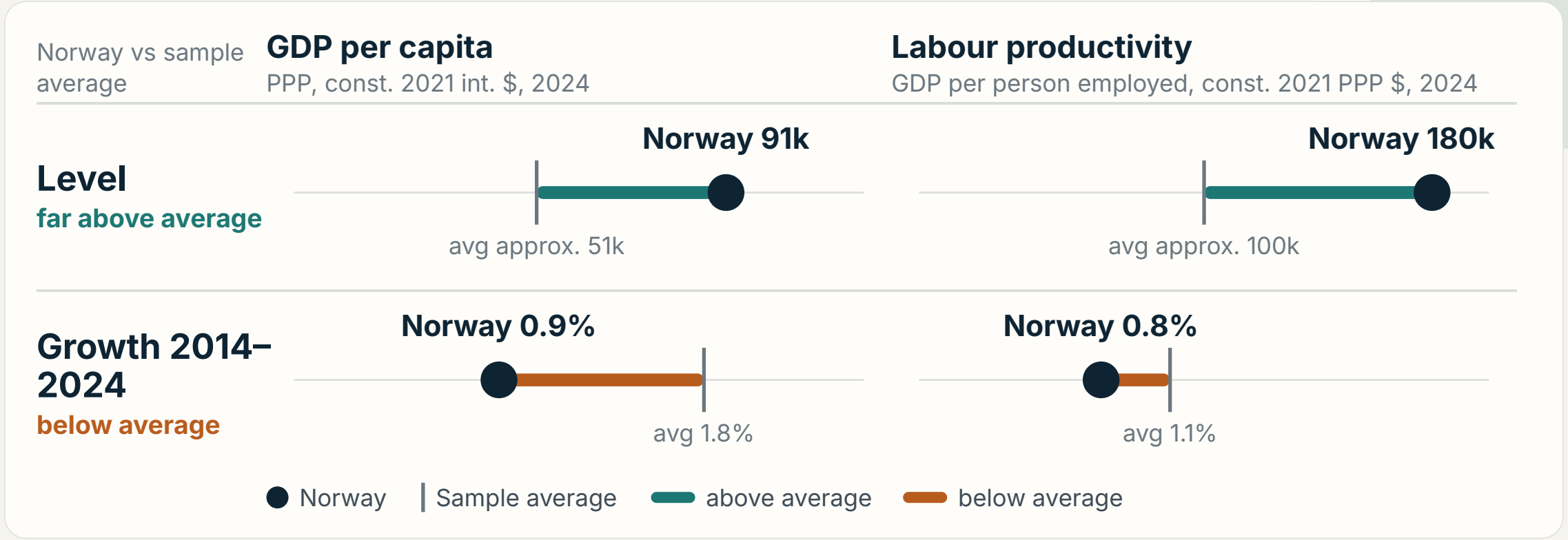
Strategy

Value proposition, ECN pipeline, recommendations, limits

RESEARCH QUESTION

How can Energy Cluster Norway help turn world-class offshore capabilities into globally competitive energy technologies?

High level, but falling behind on growth



MOC QUADRANT: HIGH BUT DECLINING

91k USD per capita · **0.9%** growth p.a. (avg 1.8%) · **100.3** USD/hour, 3rd in OECD · **57%** of goods exports petroleum

Sources: Löffler & Niemczak (2026), MOC exhibits (World Bank / ILO; positions approximate); OECD (2026); Norwegian Petroleum (2026). Figures include the petroleum sector.

Strong buffers, weak commercial output

MACRO PERFORMANCE

GDP growth

Mainland +1.7% in 2025; about +1.7% expected for 2026

Inflation

CPI 3.3% (Aug 2026), above the 2% target

Policy rate

Raised to 4.5% in Sept 2026; floating krone

Employment

Unemployment 4.7% (labour force survey, June 2026)

Fiscal stability

2.7% of the oil fund spent in 2026 and 2027; rule is 3%

Non-oil deficit

12.6% of mainland trend GDP (2027 budget): leans on the fund

MICRO PERFORMANCE

Trade

57% of goods exports are petroleum; suppliers export ≈ NOK 200bn

FDI

Net inflows ≈ USD 8bn (2023); volatile year to year

Education

Tertiary index at 165% of the EU average

Patents, R&D

IP assets rank 30 of 40; business R&D 65% of EU

Clusters

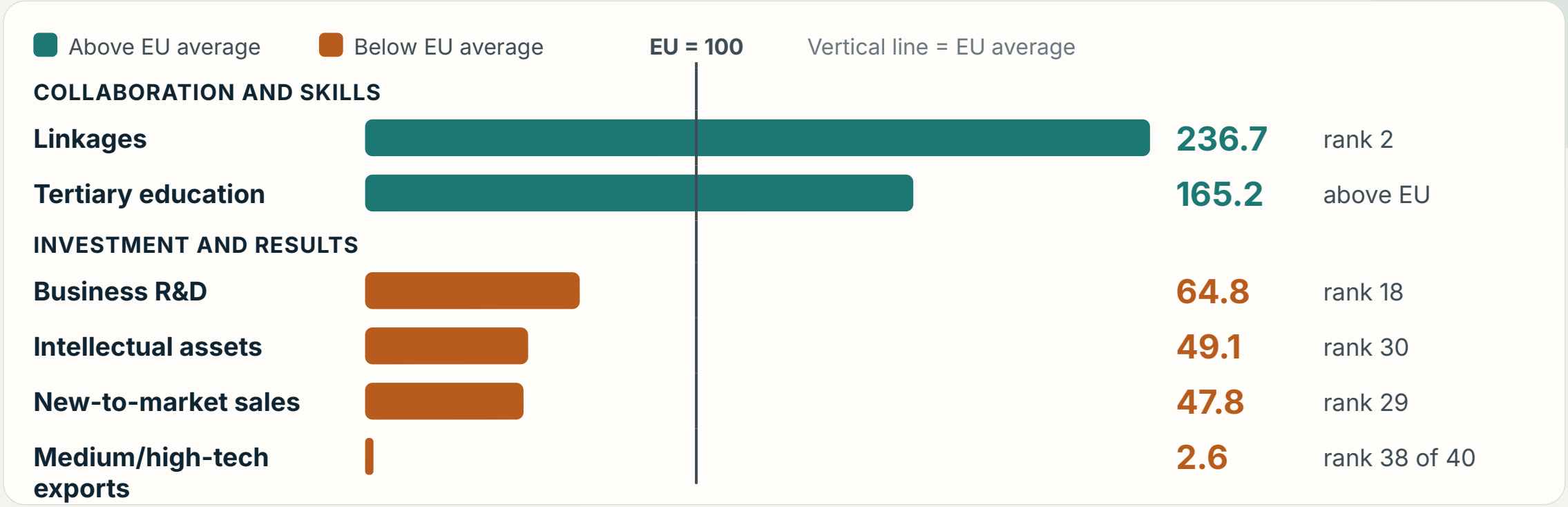
Mature NIC programme; ECN unites three clusters, 350+ members

Survey data

IMD 2026: rank 18 of 70, down six places

Data as of November 2026. Sources: Statistics Norway (2026); Norges Bank (2026); Ministry of Finance (2026a, 2026c); IMD (2026); Norwegian Petroleum (2026); Norwep (2025); World Bank; EC (2026b).

Strong in collaboration, weak in commercial results



> Knowledge is created and shared, but rarely converted into protected assets, new products and high-tech exports (rank 38 of 40).

Source: European Commission (2026b), European Innovation Scoreboard 2026, country profile Norway. Overall: Strong Innovator, 115.7% of EU, rank 10. Output shares are depressed by the petroleum-heavy export base.

Legacy, geography and macro policy

LEGACY: WHAT NORWAY INHERITED

Culture, religion. Lutheran, egalitarian, high-trust; communal work (dugnad).

Language, society. Small population (5.6m), one language, Nordic culture.

Institutions. Nordic model: rule of law, state-employer-union cooperation, 67% state share in Equinor.

Resources. Hydropower, shipping, fisheries; oil since Ekofisk (1969).

REGIONAL POSITION AND INTERACTION

Geography. Remote, high-cost north-western periphery; harsh North Sea.

Market access. Outside the EU, but in the EEA (1994), Schengen and NATO.

Energy links. Gas supplier to Europe; CO_2 storage for NL, DK and SE.

Neighbours. UK, DE, NL, DK: customers, partners and rivals.

MACRO POLICY: A STRENGTH, NOT A CONSTRAINT

Monetary policy

Inflation targeting and floating krone since 2001 (target 2% since 2018); rate raised to 4.5% in 2026 as inflation stayed above target.

Fiscal rule, stabilisation

Spend only the fund's expected real return (3% since 2017; 2.7% in 2026 and 2027), with room for countercyclical spending.

Currency, wealth buffer

The krone weakens when oil falls, helping non-oil exporters; the fund keeps oil income out of the economy.

Sources: Own assessment; Statistics Norway; Norges Bank (2026); Ministry of Finance (2026a, 2026c); Equinor (2026). EEA = European Economic Area.

Micro policy and the national Diamond

POLICY AREA	CHOICE	RATING
Structural, liberalisation	EEA market access; open trade; 78% marginal petroleum tax.	**++ open**
Competition	Open licensing rounds, but a few large buyers dominate.	**± concentrated**
Privatisation	Equinor listed in 2001; state keeps 67%.	**± state-led**
Education, R&D	Tertiary education at 165% of EU; SkatteFUNN R&D tax credit.	**+ strong**
Cluster policy	Norwegian Innovation Clusters; regional bodies merged into ECN.	**+ in place**
Demand policy	State funds Longship, Hywind Tampen (via Enova) and Utsira Nord.	**± state-funded**

NATIONAL DIAMOND

Factor conditions ++

Educated workforce, hydropower, capital; high wages.

Demand +

Sophisticated buyers, strict HSE; small home market (5.6m).

Related industries ++

Dense supplier, maritime and research networks; regionally fragmented.

Strategy and rivalry ±

High trust, but few large firms; low business R&D.

Rich but narrow: slow to create new industries.

Sources: Innovation Norway (n.d.); Equinor (2026); Norwegian Petroleum (n.d.); Ministry of Energy (2026); EC (2026b); Porter (2008). ++ strong, + good, ± mixed.

Five phases: from discovery to energy transition

1960s–70s

Foundations

Ekofisk (1969) starts production in 1971; the state creates Statoil and the Petroleum Directorate (1972) to build national control and competence.

1980s–90s

Capability building

Norwegian yards and engineering firms learn offshore work; CO_2 storage starts at Sleipner (1996); harsh-environment skills mature.

2000s

Going global

Subsea suppliers such as Aker Solutions, Subsea 7 and Kongsberg win international markets; Statoil is listed in 2001; first Hywind demonstrator (2009).

2014–20

Shock and reset

The 2014 oil-price fall forces cost cuts and layoffs; suppliers look to offshore wind and CCS; decisions on Hywind Tampen (2019) and Longship (2020) show new paths.

2021–26

Transition and consolidation

Hywind Tampen (2022/23) and the first CO_2 stored by Northern Lights (2025); ECN launches in August 2026 as one national platform.

PATTERN **Each phase added a capability; collaboration institutions came last.**

Sources: Norwegian Petroleum (n.d.); Equinor (2019, n.d.-a, n.d.-b, n.d.-c); Gassnova (2022); Northern Lights JV (2026). Phases are our own grouping.

Offshore energy technology and services

SPECIALISED SUPPLIERS

Aker Solutions, TechnipFMC, Subsea7, Kongsberg; vessels, digital and AI services, SMEs

DEMAND

Shelf operators; European emitters buying CO_2 storage; utilities and industry

Core cluster

Oil and gas · floating and fixed offshore wind · CO_2 transport and storage · offshore electrification · decommissioning

Operators: Equinor (67% state), Aker BP, Vår Energi. Flagships: Hywind Tampen, Northern Lights

KNOWLEDGE AND TALENT

NTNU, UiB, UiS, NHH, UiO; SINTEF, NORCE, IFE; test sites such as METCentre

RELATED CLUSTERS

Maritime, hydropower and grid, process industry, aquaculture, ICT

IFCs

Energy Cluster Norway (GCE Ocean Technology + Energy Transition Norway + Energy Valley) · Offshore Norge · Norwegian Offshore Wind · Norwep · Innovation Norway.

Government and regulators: ministries, Havtil, Gassnova, Enova

Own illustration in the format of Porter's cluster maps (Porter, 2008); selection of actors illustrative, not exhaustive.

A large, export-oriented cluster in transition

NOK 200bn

energy-supplier exports, 2025 (forecast; ~20% offshore wind)

350+

member companies in ECN (2026)

67%

state share in Equinor, the largest buyer (2026)

1.5 → 5+ Mt

Northern Lights CO_2 capacity a year, phase 1 (2025) → phase 2

67k → 69k

direct petroleum jobs, 2014 → 2025 (est.), after the first-ever fall in 2014–18

≈ 17%

of supplier revenue outside petroleum (2023; total NOK 532bn)

READING **Strong export base, but only a sixth of revenue is outside petroleum.**

Sources: Norwep (2025), export statistics; Energy Cluster Norway (2026); Equinor (2026, n.d.-b); Norwegian Petroleum (n.d.), employment; supplier-industry study 2023 (Multiconsult / Vista Analyse); Norwegian Offshore Wind (2026).

One national platform, three equal regions

Launched August 2026

350+ members

Equal ownership: ⅓ per region

Fully integrated by end-2027

Bergen / Vestland

GCE Ocean Technology

Subsea and ocean technology, CO_2 terminal at Øygarden: deep tech and scale-up

Stavanger / Rogaland

Energy Transition Norway

Operators, supplier networks, Utsira Nord: demand and first customers

Oslo region

Energy Valley

Headquarters, finance, digitalisation and AI: capital and internationalisation

FROM OUR CEO INTERVIEW

Model built bottom-up from interviews with 100+ companies. Services are distributed, not duplicated: MIT scale-up programme in Bergen and Boston, Technical Committee in Stavanger, open to all members. Next milestone: a secure funding model by spring 2027.

> *"It has been a key principle that all regions of the three owner clusters should be equal partners."*
> **Owe Hagesæther, CEO of Energy Cluster Norway**

Sources: Energy Cluster Norway (2026); Energy Valley (2026); O. Hagesæther, CEO of ECN, written interview, 25 October 2026.

Strong inputs, weak demand for new technology at scale

Factor conditions

++ · stable

World-class subsea engineering, test sites and research; scarce specialists; grid limits

Demand

+ · CCS rising

Very sophisticated in oil and gas; for CCS and floating wind still reliant on state support

Related industries

++ · rising

Maritime, power and process industries recombine; ECN reduces regional fragmentation

Strategy and rivalry

± · stable

Few powerful buyers (Equinor 67% state-owned); low business R&D; scale-ups stay project-based

THE WEAK LINK

Buyer concentration and state-backed projects decide whether a demonstration ever becomes a commercial order.

Own assessment. Strength: ++ strong, + good, ± mixed. Based on European Commission (2026b), Equinor (2026), Bjørndal et al. (2025).

Strategic issues and policies already in place

ISSUE	POLICY IN PLACE	REMAINING GAP
Demand Few buyers; pilots rarely become orders.	State aid: Utsira Nord (cap NOK 35bn), Enova support for Hywind Tampen, Longship CCS.	No lead customer or procurement route built into support.
Strategy and rivalry Operator concentration; business R&D at 65% of EU.	SkatteFUNN tax credit; open licensing rounds; state ownership of Equinor.	Incentives reward R&D spend, not commercial conversion.
Factor conditions Scarce specialists, grid limits, IP output (rank 30).	Research Council programmes (e.g. Green Platform); universities; test sites such as METCentre.	Test infrastructure and skills not coordinated across regions.
Related clusters and IFCs Regional fragmentation of cluster bodies.	Norwegian Innovation Clusters programme; ECN merger in 2026.	No common gate process or outcome metrics yet.

Sources: Own assessment; Innovation Norway (n.d.); Ministry of Energy (2026); Equinor (n.d.-b); European Commission (2026b).

Validation alone does not create scale

Hywind

floating wind

2009

PILOT

Demonstrator off Karmøy, 2009

Committed buyer

turning point

2022/23

FIRST CUSTOMER

Hywind Tampen powers Equinor's Snorre and Gullfaks fields (2022/23)

Utsira Nord

NEXT STEP

Utsira Nord: state-aid competition, aid capped at NOK 35bn

LESSON Still policy-dependent: terms revised in 2026 over an immature supply chain

Northern Lights

CO_2 storage

2020

PILOT

Longship CCS, investment decision 2020

Committed buyer

turning point

2025

FIRST CUSTOMER

State-supported Norwegian emitters; first CO_2 stored 2025

Phase 2

NEXT STEP

Contracts in NL, DK and SE; phase 2 from 1.5 to 5+ Mt CO_2 p.a.

LESSON Anchor customers plus an export market turned a public project into a service

Both paths moved forward only once a committed buyer and a market route were in place.

CONTEXT CO_2 storage under the seabed has been proven at Sleipner since 1996. A market only emerged once a buyer was built in (Longship, 2020).

Sources: Equinor (2019, n.d.-a, n.d.-b, n.d.-c); Regjeringen (2019); Gassnova (2022); Ministry of Energy (2026); Northern Lights JV (2026); offshoreWIND.biz (2026).

Three North Sea locations compete for the same role

Aberdeen / Scotland

Oil and gas supply chain closest to Norway's; ScotWind 2022 leased 24.8 GW, of which 14.6 GW floating.

Esbjerg / Denmark

Europe's leading port for shipping out offshore wind; Greensand Future CO_2 storage approved in Dec 2025.

Rotterdam / Netherlands

Europe's largest industrial port; Porthos to store 2.5 Mt CO_2 a year from 2026/27.

NORWAY'S EDGE

- ✓ Cross-border CO_2 storage in operation since 2025 (Northern Lights).
- ✓ Floating wind powering platforms since 2022/23 (Hywind Tampen).
- ⚠ **Weak spot: higher costs, small manufacturing base.**

IMPLICATION **Compete on proven harsh-environment systems, not on volume.**

Sources: Crown Estate Scotland (2022); Port Esbjerg (n.d.); Danish Energy Agency (2025); Porthos (n.d.); Northern Lights JV (2026); Equinor (n.d.-a); weak spot: own assessment.

Norway as Europe’s reference location for offshore energy systems in harsh environments

The place where offshore energy technologies are proven, scaled and exported: from oil and gas to floating wind, CO_2 storage and offshore electrification.

WHAT NORWAY SHOULD NOT DO

- No subsidy race for mass manufacturing without a cost or scale advantage
- No duplicate test infrastructure across the three regions
- No permanent operating subsidies without a credible path to competitiveness

One national pipeline from industrial need to first order

GATE 1	GATE 2	GATE 3	GATE 4	GATE 5	GATE 6
Verified industrial need	Capable consortium	Technical validation	Demonstration	First commercial order	International scale-up

Before the demonstration gate: a named lead customer, purchasing criteria and a procurement route.

1 · Convert pilots into first commercial orders

Common six-gate process; measure conversion, not activity

2 · Build one national commercialisation pipeline

Shared gates and stop/continue decisions across regions; challenge roadmaps without pre-selected technologies

3 · Scale successful technologies internationally

Norwep, export finance and North Sea partners on board before the demonstration ends

WHO DECIDES?

Funders link co-funding to the gates; ECN prepares reviews with its Technical Committee and independent experts. Gates advise, funders decide.

WHY COMMIT EARLY?

The first unit is co-funded, risk is shared and purchasing criteria are set with suppliers.

WHO PAYS FOR ECN?

Gate reviews become a member service in the funding model due in spring 2027.

Stage-gate logic (Cooper, 1990) applied at cluster level; O. Hagesæther, written interview, 25 October 2026.

Recommendations: who needs to do what

National government

- Turn support into demand: a multi-year, technology-neutral procurement calendar for low-emission offshore energy.
- Tie aid to a named lead customer and a falling support path.

Regional and local government

- Fast-track ports, land and grid connections in Bergen, Stavanger and Øygarden.
- Share, not duplicate, test sites and skills programmes across the regions.

Private sector

- Operators act as lead customers and publish purchasing criteria for first units.
- Suppliers raise business R&D, protect IP and co-fund demonstrations.

Universities and institutes

- Reward IP, spin-outs and industrial PhDs, not only publications.
- Open test sites and train subsea, wind and CCS specialists.

Trade associations

- Bundle member demand into joint buyer roadmaps (Offshore Norge, Norwegian Offshore Wind).
- Lead export promotion with Norwep and Innovation Norway.

Collaboration bodies (ECN)

- Run the six-gate pipeline; funders link co-funding to the gates.
- Report conversion: pilot-to-order rate and export revenue.

Source: Own recommendations, derived from the Diamond analysis and the Hywind and Northern Lights cases.

Measure conversion, not activity

0–12 months

Governance for 2027, challenge map, gate criteria, portfolio baseline

Proof: shared data model, baseline conversion data

12–36 months

Challenge-based calls, cross-regional consortia, lead customers, co-funded demonstrations

Proof: more projects with purchasing criteria and private co-finance

36–60 months

Scale what works, repeat procurement, target-market partnerships, stop weak projects

Proof: first deployments, repeat orders, export contracts

Pilots into orders

Share with named lead buyer; pilot-to-order rate

One pipeline

Time from need to pilot; private funding per public NOK

Scale-up

Repeat orders; export revenue; EIS output indicators

Shared value

Verified emission savings; supplier growth; skills

CREATING SHARED VALUE

Subsea CCS: Northern Lights sells CO_2 storage as a service: exports and decarbonisation at once.

Floating wind: Hywind Tampen powers oil and gas platforms: lifecycle services and lower emissions.

Sources: Own analysis based on Porter & Kramer (2011); Northern Lights JV (2026); Equinor (n.d.-a).

Norway has the knowledge and the collaboration. What it needs is a reliable path from pilot to first order.

Diagnosis

High but declining; a commercialisation gap

Bottleneck

Demand for new technology at commercial scale

Role of ECN

One pipeline, shared gates, outcome-based measurement

LIMITS EIS output shares partly reflect the petroleum export base · two state-backed cases · one interview with ECN itself. The final report adds cluster statistics and a second interview.

Thank you. We look forward to your questions.